

Affordable Next-Generation Business Continuity with Intel and Microsoft

How to combine virtualization and related technologies from Intel and Microsoft to save money while achieving superior availability and disaster recovery

White Paper

**Affordable Next-
Generation Business
Continuity**

**Combining Virtualization
and Related Technologies
from Intel and Microsoft**

Abstract

Robust business continuity plans, which include high availability, data protection, and disaster recovery, are vital. Regardless of where you are along the continuum of infrastructure optimization—Basic, Standard, Rationalized, or Dynamic—you can boost the effectiveness of your business continuity plans through virtualization technology, all while reducing the costs. Virtualization products from Microsoft, such as Hyper-V™ technology in Windows Server® 2008 and Microsoft® System Center management tools, powered by the latest Intel® Xeon® processors with Intel Virtualization Technology, can help you reach the next level of business continuity: an optimized infrastructure that provides enhanced availability, data protection, and disaster recovery – all at a lower total cost than prior business continuity solutions.

Microsoft®



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Effective Business Continuity Planning: Modern Solutions to Past Challenges

The need for effective business continuity plans is greater than ever. Any downtime costs money, and data loss can be catastrophic. Hardware and software failures, natural disasters, and planned maintenance, all result in downtime that can range from an inconvenience to a situation that can bring business to a halt. Not only do you have frustrated end users and overtaxed IT departments, you might also end up with damaged brand equity and the loss of critical information and revenue.

Unfortunately, implementing an effective business continuity plan has often been time-consuming and expensive. For example, traditional business continuity solutions required you to acquire and maintain extra equipment (usually offsite) that mirrored your production environment down to the hardware level. This doubled workload meant a very high total cost of ownership (TCO). Because of the cost and difficulty, many organizations simply don't have comprehensive business continuity or disaster recovery plans to cover all their devices, data, and applications.

Because of the cost, difficulty and time required, many IT organizations simply don't have comprehensive business continuity or disaster recovery plans to cover all their devices, data, and applications. The answer is virtualization.

A solution to these complexity and cost issues is now possible with virtualization technology. Virtualization has become much more than just server consolidation. With virtualization, you can realize numerous business continuity benefits, including

- Lower TCO
- Improved flexibility
- Higher reliability
- Quicker backups
- Faster data recovery

Virtualization can provide these benefits by making it easier and less costly to reach higher levels of system availability, as well as superior backup and recovery. This is extremely important as companies strive to achieve greater results with less expenditure.

Building a virtualization-based business continuity solution using proven, complementary technology from Intel and Microsoft can lead to a substantially reduced total cost of ownership (TCO). But saving money is only the beginning. Virtualization also enables enhanced high availability and disaster recovery for your mission-critical systems.

The Four Levels of Core Infrastructure Optimization (Core I/O)

Virtualization-based business continuity solutions complement the dynamic datacenter vision shared by Microsoft and Intel. To help businesses move toward a vision of automated dynamic systems and applications, Microsoft has created the Core Infrastructure Optimization Model. This model defines four levels of automation in the computing platform, each of which has specific and identifiable characteristics. The four levels of optimization start with Basic, in which the infrastructure has no disaster recovery or backup plan; then move in stages up to Dynamic, in which continuous data protection and service availability guarantee business continuity.

The levels of optimization are as follows:

Basic (IT is a cost center):

- Backup and restore functionality isn't present for most critical servers.
- Regulatory compliance requires more effort due to lack of archive and backup services.
- No disaster recovery plan is in place.
- Data is stored in random locations with no network backups.

Standardized (IT is an efficient cost center):

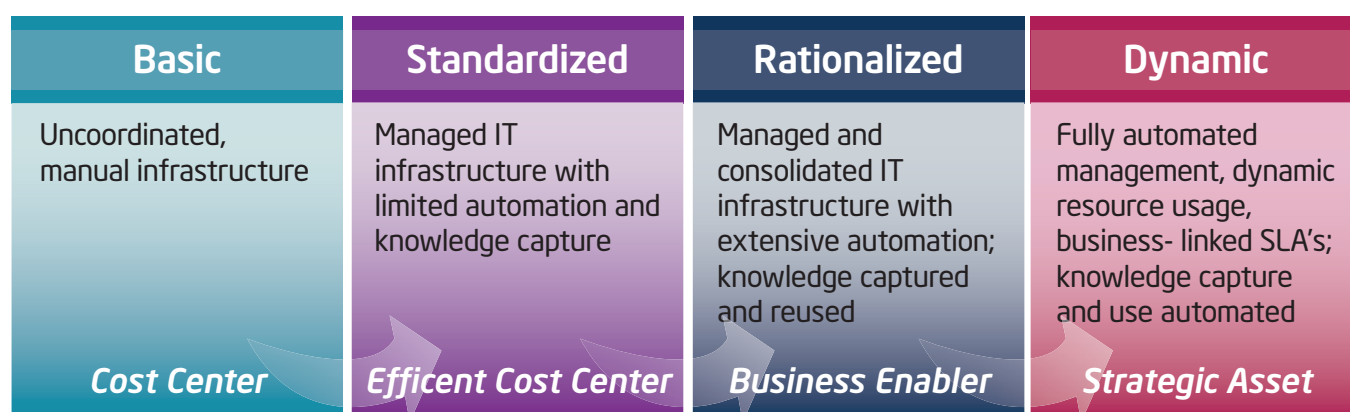
- Backup and restore is available for critical servers.
- Backups are at the workgroup level only.
- Some automation of user state migration is available.

Rationalized (IT is a business enabler):

- Backup and restore is available for servers with Service Level Agreements (SLAs).
- Backup and restore efficiency is enhanced through virtualization technology.
- Backup is managed at the company level.
- User state is preserved and restored for deployment.

Dynamic (IT is a strategic asset):

- Backup and restore exists for all servers with SLA and for desktop data.
- Continuous real-time data protection exists for servers and desktops, including user states, in warm- and hot-failover environments.
- Guarantees business continuity and high availability.
- Rapid and reliable recovery is available.
- Regulatory compliance becomes possible.



It is now possible to diagnose and resolve problems even if the operating system is not responding, and to monitor system health even if management agents are missing on the target PC. In addition, IT can access critical system information even if hardware (such as a hard drive) has failed.

Virtualization and the Core Infrastructure Optimization Model

As an organization moves toward the Dynamic level of infrastructure optimization, benefits are realized at each step of the way. However, it is important to note that the ideal position on this continuum for each organization is dependent on a wide range of factors (for example, costs and unique business processes). Nevertheless, virtualization increases the number of options available to IT organizations as they determine where on this continuum they should strive to be.

Improving Business Continuity through Virtualization

Virtualization technology can be crucial in helping organizations optimize their IT infrastructures. Through virtualization, organizations at the Basic optimization level can substantially improve resource utilization by taking advantage of server consolidation, and can realize power-saving goals. Virtualization can assist in application testing, staging, and moving workloads into production. While IT may still be viewed as a cost center, virtualization can help that center become much more efficient and thus contain costs.

Virtualization, TCO, and CAPEX

By decoupling the operating system from the physical hardware while providing the inherent benefits of virtual machine isolation and portability, virtualization allows organizations to extract greater utilization and value from their hardware investments.

Thus, moving to a virtual machine model can lead to lower TCO and capital expenditures (CAPEX), freeing up valuable dollars that can be invested into new IT projects or simply used to improve the bottom line. Moreover, virtualization will take organizations down the path of creating a more dynamic enterprise.

Prior to virtualization, the growth in x86 server and desktop deployments led to IT infrastructure and operational challenges, such as:

- Low infrastructure utilization
- Increasing physical infrastructure costs
- Increasing IT management costs
- Insufficient and increasingly costly failover and disaster protection
- High-maintenance end-user desktops

By using virtualization to convert physical servers into virtual machines, and by running the virtual machines on fewer, upwardly scalable and more reliable enterprise-class servers, organizations have been able to realize several significant business benefits:

- **Lower TCO.** Consolidation lowers hardware, facility, power, cooling, and management costs. Intel Xeon processors with Intel Virtualization technology further decrease TCO by supporting more instructions per clock cycle.
- **Optimized infrastructure.** Businesses gain significant advantages in asset utilization, reliability, and availability.
- **Improved flexibility.** IT administrators can freely integrate 32-bit and 64-bit workloads in the same environment, balance workloads, isolate applications, and rapidly respond to demands for more or different servers and applications.
- **Automated energy efficiency.** Within a single server, Intel® Intelligent Power Technology minimizes power consumption based on workload.

Reducing TCO and CAPEX are powerful justifications to begin introducing virtualization into your IT infrastructure. However, these benefits can be further magnified when coupled with Intel® server processors. Intel's commitment to virtualized infrastructures has led to a set of products and technologies that help organizations to maximize the performance, reliability, and efficiency of the hardware powering their virtual infrastructure.

Using Intel Xeon processors in your virtualization scenarios makes sound economic sense. Compared to single-core servers, new Xeon processor-based servers enable

- Up to 9 times the performance per server
- Up to 90% lower operating costs

Through this technology, organizations can reduce power, cooling, and real estate costs while improving performance—all with a lower operating cost. But using virtualization to consolidate servers is only the beginning to what this technology can do.

Taking Virtualization to the Next Level

Forward-thinking organizations have recognized that virtualization can do more for IT than just consolidating servers and isolating applications. They have discovered virtualization to be a key to further efficiencies and cost savings. This next level of virtualization represents the expanding capabilities of virtualized platforms to help optimize existing IT infrastructures to improve business continuity, and to provide solutions for high availability, data protection, and data recovery.

Increasingly sophisticated hardware and software solutions are extending virtualization implementations beyond simple server consolidation to include new business continuity scenarios. These virtualization technology solutions offer a host of benefits specific to business continuity solutions, such as:

- **Lower power, space, and cooling costs.** Server consolidation through virtualization technology results in more efficient data center operations with more applications running on fewer physical machines. Intel Xeon processors, with advanced power management features supported by Windows Server 2008, further reduce power and cooling requirements.
- **Improved manageability.** Virtualization enables optimal resource allocation decisions that deliver the highest quality of service available. IT administrators can easily provision new virtual machines for development teams and business units, reducing delivery time from weeks or months to days or hours. The inherent portability of virtual machines makes it easy to move from one machine to another.

- **Accelerated ROI.** Reduces the number of servers and improves overall utilization of hardware and software resources. Depending on workload requirements, servers running virtual machines can operate at 60 percent utilization or higher. Increased utilization leads to more effective use of server resources while reducing the footprint in the data center. Virtualization also maximizes the value of existing and future hardware investments, as virtual machines themselves are hardware independent.
- **Reduces the number of potential failure points.** By reducing failure points, virtualization frees up CAPEX to create hardware redundancy initiatives.
- **Increased uptime.** Virtualization technologies enable IT organizations to increase uptime with dynamic response policies that can transparently move workloads to other machines when triggered by critical events.

Traditional High Availability Solutions

Broadly speaking, high availability can be defined as the implementation of a system design that ensures a high level of operational continuity, even during hardware failure or interruption. Availability is somewhat subjective: Availability requirements will always center on individual business needs in a given situation. The common element is the need to minimize the time workloads are unavailable.

Downtime comes in two flavors: planned and unplanned. Planned downtime is the result of management-initiated events, such as maintenance. By contrast (and by definition), unplanned downtime is the result of events not within direct control of IT. Such events range from the minor (e.g., corrupted hardware drivers or a failed NIC) to the catastrophic (e.g., flood, fire, or earthquake).

Regardless of the cause, downtime is costly. A heavily used e-mail or database server can easily cost a business thousands or tens of thousands of dollars for every hour that it is unavailable. Beyond the benefits and advantages brought to an organization by an IT solution, technology and business decision-makers should also think about how to provide business continuity during the inevitable downtime inherent in these solutions.

Traditional high availability solutions have involved creating and maintaining identical computing environments (including hardware, operating systems, applications, and data), and clustering these identical environments together with software that monitors their health. When the primary environment experiences downtime, the secondary environment takes over, providing a seamless experience for end users. While effective and time-tested, these traditional solutions are expensive and complex to implement and maintain.

As a result, only the largest of organizations could afford such a solution. Most organizations have been left to cobble together solutions that, while affordable, offer less-than-adequate business continuity.

High Availability and Virtualization

Due to the costs and complexity of traditional high availability solutions, an increasingly popular solution for improving business continuity is virtualization. Virtualization brings high availability to a broader set of mission-critical systems, and does so in a more efficient and cost-effective fashion. Because virtual machines are abstracted from the hardware, you can repurpose existing servers for business continuity or disaster recovery. While some hardware requirements must be met, such as processors equipped with hardware virtualization, servers in your recovery site do not have to be identical to those in your primary operations site. Perhaps even more important, virtualization can maximize your hardware investment by introducing fewer, yet more powerful, servers. These servers can be equipped with the latest in Intel Xeon processor-enabled performance and virtualization enhancements, allowing you to port virtualized workloads to these faster and more energy-efficient machines with relative ease.

Virtualization enables you to maintain a nearly instantaneous failover plan that provides business continuity in spite of disruptive events.

With virtualization, you can enable automated backup, replication, and rapid movement of servers, desktops, and applications in response to untoward events.

Virtualization enables you to maintain a nearly dynamic failover response plan that provides business continuity in spite of disruptive events. With virtualization, you can enable automated backup, replication, and rapid movement of servers, desktops, and applications in response to untoward events.

Getting the Most Out of Virtualization with Microsoft Technologies Running on Intel Xeon Processor-Based Servers

Introduction to Microsoft Virtualization Tools and Related Technologies

Microsoft provides a comprehensive set of tools for implementing and managing complex virtual infrastructures. Such a toolset allows organizations to design and roll out virtualization-based

business continuity solutions. The following Microsoft products and technologies can all play a role in such a business continuity solution:

- **Windows Server® 2008 Hyper-V®.** Provides a virtualization platform upon which virtual machines can run.
- **Failover Clustering and Quick Migration.** Provides high availability and scalability for servers and applications.
- **Microsoft® System Center Data Protection Manager.** Provides live backup of virtual machines to support business continuity.
- **Microsoft® System Center Operations Manager.** Allows advanced monitoring and service reporting for consolidation, configuration, utilization, and growth projections.
- **Microsoft® System Center Virtual Machine Manager.** Enables centralized management of the virtual infrastructure.

Windows Server 2008 Hyper-V

Hyper-V, Microsoft's hypervisor solution, is a thin layer of software that sits between the hardware and the operating system. It allows multiple unmodified operating systems to run simultaneously on a host server. It also enables simple partitioning and maintains strong isolation between partitions.

Hyper-V has been designed to run on servers with hardware-assisted virtualization technology, and it takes full advantage of hardware-assisted virtualization, such as that provided by the Intel® Virtualization Technology suite, or Intel VT.

Intel VT is a set of enhancements to Intel processors, chipsets, and I/O devices that reduces virtualization software overhead and enables more powerful and efficient virtualization solutions.

Hyper-V is an ideal software complement for an Intel Xeon processor-based server with Intel VT. Hyper-V overcomes many of the challenges faced by previous software only-based virtualization technologies by offering the following advantages:

- **Reduced complexity.** Hyper-V supports non-Microsoft and legacy virtual machines, including Windows®, Linux, and Xen-enabled Linux, without modifying the operating systems.
- **Improved performance.** Virtual machines can use larger amounts of physical resources, including up to 64 GB of memory per virtual machine. Hyper-V also supports pass-through disk access and will allow up to four virtual processors to be assigned to each virtual machine. Further, Hyper-V uses page sharing to optimize memory by maintaining single copies of the identical memory pages shared by multiple virtual machines, thus reducing resource utilization on the physical machine.

- **Increased functionality.** With Hyper-V, a single physical machine can simultaneously run virtualized 32-bit and 64-bit operating systems. In addition, Hyper-V offers advanced networking capabilities for each virtual machine, including VLAN, network address translation, firewall, the ability to quarantine individual virtual machines, and the option to add or remove virtual NICs without downtime.
- **Improved reliability and protection.** Hyper-V uses the clustering functionality in Windows Server 2008 to provide host and guest clustering for failover and load balancing across LANs and WANs. IT administrators can cluster multiple physical servers all running virtual machines to minimize the impact of one server failing.
- **Simplified management.** Hyper-V allows quick migration to migrate loads from one node to another in a cluster for easy maintenance and management. Quick migration can be used not only within a subnet but across your entire network, extending this capability to a wider geography.

These features alone provide a high degree of flexibility and power for virtualization. When coupled with a foundation of Intel Virtualization Technology and Intel Xeon processors, they present a powerful combination for complex virtualization strategies.

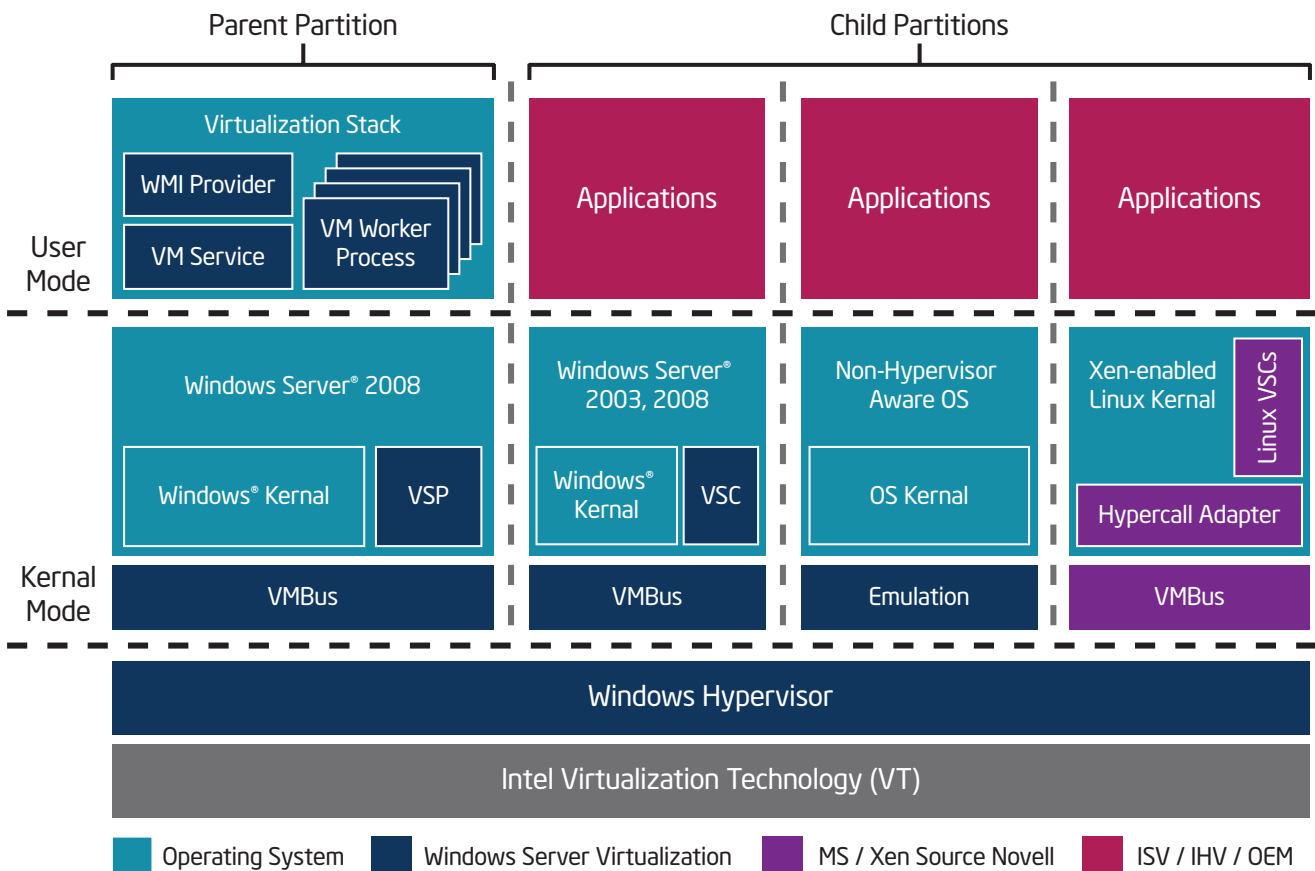
Implications for Business Continuity

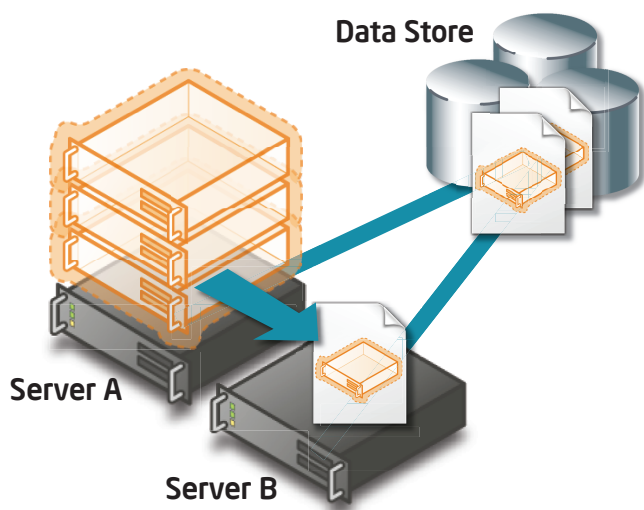
Since Hyper-V provides the platform for running virtual machines, it sets the stage for virtualization-based business continuity. By giving organizations the ability to run their critical workloads as virtual machines, Hyper-V offers all the advantages inherent to virtual machines to these workloads – hardware abstraction, isolation, and portability. These key benefits provide the foundation for business continuity.

Failover Clustering and Quick Migration

Failover clustering is an important feature of the Windows Server platform that can improve availability. When one node fails, another node begins to provide service in its place. This process is called failover. Failover clustering with Windows Servers is available only with Windows Server 2008 Enterprise and Datacenter editions.

Windows Server 2008 failover clusters provide high availability and scalability for mission-critical applications such as databases, messaging systems, file and print services, and virtualized workloads. Multiple servers (nodes) in a cluster remain in constant communication. If one of the nodes in a cluster becomes unavailable (such as a result of a hardware or software failure or having been taken down for maintenance), another node immediately begins providing service.





Users who are accessing a service continue to access the service and are unaware that it is now being provided from a different server (node).

When planning a highly available Hyper-V deployment, Windows Server 2008 failover clustering is one of the primary tools used to achieve a fault-tolerant design. There are significant improvements in Windows Server 2008 failover clustering that make configuring cluster environments more streamlined and easy to manage.

By combining Windows Server 2008 failover clustering with Hyper-V, Microsoft enables organizations to implement a mature, time-tested technology for empowering high availability in a virtual infrastructure. Moreover, many enterprise administrators are already familiar with server clustering technology in Windows operating systems, and thus current skill sets can be applied to deploy the solution.

Windows Server 2008 failover clustering supports quick migration of virtual machines. With quick migration, you can rapidly migrate a running virtual machine from one physical host system to another within a subnet or across your network with minimal downtime, taking advantage of the familiar high-availability capabilities of Windows Server and Microsoft System Center management tools. Using Hyper-V and quick migration, you can consolidate physical servers and, at the same time, maintain the availability and flexibility of mission-critical services during scheduled maintenance and quickly restore services after unplanned downtime.

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Implications for Business Continuity

Failover clustering provides a fault-tolerant infrastructure for both the physical hosts and virtual machines. When combined with Hyper-V, this can be extended to virtual machines to provide Quick Migration functionality, allowing virtual workloads to be moved quickly and easily between physical host servers.

System Center Data Protection Manager

Microsoft® System Center Data Protection Manager (DPM) is designed to provide disk-to-disk-to-tape protection for infrastructures that depend on Microsoft virtualization technology.

DPM is capable of protecting virtual machines without downtime. Using shadow copy-based block-level protection of your virtual disks, DPM delivers fast backup and efficient compression that optimizes use of disk space. DPM's agent model supports protecting all virtual machines within a host through a single agent on the host platform; alternately, agents can be installed within each virtual instance for an extended range of protection and recovery options.

Unlike backup technologies from other virtualization vendors, DPM protects physical and virtual servers running Microsoft® SQL Server, Exchange® Server, and SharePoint® products and technologies.

Data Protection Manager supports virtualization solutions through Live Backup and VM Snapshots, which offer the following benefits:

- **Live Backup.** The Volume Shadow Copy Services feature enables backup of a complete virtual machine, including data and the virtual machine state, without downtime. This simplifies virtual machine management and provides a smooth path to robust business continuity.
- **VM Snapshots.** This tool allows administrators to save the virtual machine's state, providing the ability to reset the machine to a previous "known good" state. Snapshots are commonly used before applying upgrades or patches, but can also be used to create a safeguard copy of the virtual machine before making any configuration changes.

Implications for Business Continuity

DPM should be considered a key component of a virtualized business continuity plan. The ability to back up live virtual machines with DPM means there is no downtime when backing up key business workloads, such as virtualized databases and mail servers. DPM also enhances business continuity with its unique VM Snapshot capability, which enables IT to quickly roll back from a possible update misadventure. And in the event that an entire disk should fail, DPM enables for high-speed restore using its disk-to-disk restoration routine.

System Center Operations Manager

Microsoft® System Center Operations Manager (SCOM) enables administrators to graphically define the components that make up a service. Once the service model is defined, services are managed like any other device. Events can be monitored, performance can be tracked, and overall health can be viewed and reported on. Services can also be combined to allow monitoring of more complex services that are made up of different sub-services.

SCOM monitors both physical and virtual machines and helps automate event responses. It identifies important but non-urgent alerts and can apply filtering rules that trigger workflow processes (for example, sending an e-mail, generating a support ticket, etc.), which facilitates proactive management. SCOM also provides robust health monitoring, helping administrators identify virtual machines that need to be patched or upgraded to meet policy requirements; verify that applications are running correctly; check for memory leaks or conflicts; and monitor workloads, memory use, storage levels, and CPU utilization. It also provides service reporting for consolidation, configuration, utilization, and growth projections.

Implications for Business Continuity

SCOM provides critical failure notification, which helps administrators recognize and appropriately respond to failures. The reliability, availability, and serviceability (RAS) features found in Intel Xeon processors, such as support for Windows Hardware Error Architecture (WHEA), integrate with this high level of health monitoring to reduce recovery times and system crashes related to hardware errors.

In addition, memory controller features include an Error Correcting Code (ECC) system bus, new memory mirroring, and I/O hot plug to fortify disaster recovery strategies by doubling the performance and redundancy features at the hardware level.

This integrated approach for monitoring and alerting both virtual and physical environments provides the administrator a great advantage, since he can work proactively before a critical failure takes place. And if a failure takes place without warning, SCOM can immediately inform the appropriate personnel to take corrective actions. SCOM provides that “single pane of glass” that delivers the necessary insight to keep enterprise IT running and ensure business continuity.

System Center Virtual Machine Manager

Microsoft® System Center Virtual Machine Manager (SCVMM) 2008 is a comprehensive management solution for the virtualized data center, enabling increased physical server utilization; centralized management of the virtual machine infrastructure; and rapid provisioning of new virtual machines by the administrator, delegated administrator, and authorized end users.

SCVMM provides complete support for consolidating physical servers onto a virtual infrastructure. This centralized management console delivers many powerful tools to simplify virtual machine management, helping IT administrators rapidly provision, deploy, and migrate virtual machines in hours or days instead of weeks or months.

Table 1 (next page) lists some of key features and capabilities included with SCVMM related to high availability and business continuity.

Feature	Details
Host Cluster Support for “High Availability” Virtual Machines	With greatly expanded support for failover clusters, VMM improves its “high availability” capabilities for managing mission-critical virtual machines. VMM is fully cluster-aware, meaning it can detect and manage Hyper-V host clusters as a single unit. Enhanced high availability means more uptime and improved business continuity.
Intelligent Placement Analysis	With intelligent placement analysis, SCVMM analyzes the expected load of a virtual machine against available physical servers and recommends the best server for the load. Virtual Machine Manager conducts extensive data analysis using a number of factors, including using historical data provided by System Center Operations Manager.
Utilization Counters	Virtual Machine Manager includes utilization counters to help administrators effectively monitor virtual machines and Hyper-V host server performance and to inform System Center Operations Manager of potential problems before they become business continuity issues.
Physical-to-Virtual and Virtual-to-Physical Machine Conversions	Virtual Machine Manager provides physical-to-virtual (P2V) machine conversion technology that removes the need to halt the physical server. IT administrators can convert a running server to a virtual machine at disk speed, saving time and reducing complexity. Virtual-to-physical (V2P) migration is also available. P2V and V2P conversions can be used to accelerate disaster recovery in a wide number of scenarios.
Centralized Library	Virtual Machine Manager features a centralized library that organizes and manages all the building blocks of a virtualized infrastructure, including stored virtual machines, virtual hard disks, CD/DVD software images, post-deployment customization scripts, hardware configurations, and templates. Pre-defined templates and library images enable even inexperienced administrators to spin up new virtual machines in key business continuity incident response situations.
Integration with Hardware-Assisted Virtualization	As part of the Intel and Microsoft commitment to complementary virtualization solutions, Virtual Machine Manager lets administrators select hardware-assisted virtualization for each virtual machine. This simple integration method enables businesses to immediately take advantage of the virtualization benefits built into quad-core Intel Xeon processor-based servers.

Implications for Business Continuity

In addition to these features, SCVMM delivers key business continuity support in its new Performance and Resource Optimization (PRO) feature. PRO enables the dynamic management of virtual resources through Management Packs that are PRO enabled. Utilizing the deep monitoring capabilities of System Center Operations Manager 2007, PRO enables administrators to establish remedial actions for VMM to execute if poor performance or pending hardware failures are identified in hardware, operating systems, or applications. PRO is just another example of how Microsoft System Center software plus advanced WHEA support found in Intel Xeon processors have multiplicative effects when it comes to a consistent and reliable virtual business continuity infrastructure.

Future Microsoft Products and Technologies for Business Continuity

Microsoft continues its tradition of excellence with new features and functionalities that make the most of virtualization capabilities. Windows Server® 2008 R2 Hyper-V—the next generation of Microsoft’s hypervisor technology—builds on the rock-solid architecture of Windows Server 2008 Hyper-V, enabling new dynamic scenarios, such as:

- Increased server consolidation
- Dynamic data center
- Centralized virtual desktops

In addition, Windows Server 2008 R2 Hyper-V includes live migration, enhanced support for the new Nehalem microarchitecture, the ability to hot add or remove virtual disks, and improved networking functionality.

Live Migration. With Hyper-V live migration, you can move a virtual machine from one server to another without loss of service or perceived downtime. Live migration enables:

- Load-balancing VMs for power
- Load-balancing VMs for CPU performance
- Upgrade of host hardware and maintenance

In addition, Hyper-V live migration gives you greater agility, since data centers with multiple Hyper-V physical hosts can move running VMs to the best available server for performance, scaling, or optimal consolidation without impacting users. Live migration reduces costs and increases productivity, since data centers with multiple Hyper-V hosts can service those systems in a more controlled fashion, enabling scenarios such as scheduled maintenance during regular business hours. Live migration makes it possible to keep VMs online, even during maintenance, increasing productivity for both users and server administrators. Data centers will also be able to reduce power consumption by dynamically increasing consolidation ratios and powering off unused physical hosts during times of lower demand.

Dynamic Virtual Machine Storage. Windows Server 2008 R2 Hyper-V includes improvements to virtual machine storage. First, you will be able to add and remove VHDs and pass-through disks to and from a running VM without requiring a reboot. This enables:

- Storage growth in VMs without downtime
- Additional data center backup scenarios
- New SQL Server / Exchange Server scenarios

You will also be able to add and remove both virtual hard disks and physical disks to and from existing SCSI controllers of virtual machines.

Enhanced Processor Support. Windows Server 2008 R2 Hyper-V allows you to take advantage of the latest in processor innovations. Several key enhancements will help this happen:

- Support for up to 32 physical processor cores.
- Support for Intel processors' Extended Page Tables, improving memory management performance, reducing memory copies, and decreasing memory usage. This is known as Second-Level Address Translation (SLAT). SLAT adds a second level of paging below the architectural x86/x64 paging tables in x86/x64 processors, improving performance of virtual machines.

- CPU Core Parking enables Windows and Hyper-V to consolidate processing onto the fewest number of possible processor cores, and suspends any cores that are inactive.

Enhanced Networking Functionality. Windows Server 2008 R2 Server Hyper-V allows TCP/IP traffic in a VM to be offloaded to a physical NIC on the host computer, reducing CPU burden and improving performance. In addition, Live Migration is fully supported with Full TCP Offload. Virtual machines will now support jumbo frames to enhance network throughput and reduce processor utilization.

Hyper-V also supports Intel® Virtual Machine Device Queues. Virtual Machine Device Queues (VMDq) is a silicon-level technology that offloads network I/O management burden from the hypervisor. Multiple queues and sorting intelligence in the silicon support enhanced network traffic flow in the virtual environment, freeing processor cycles for application work. This improves efficiency in data transactions toward the destined VM, and increases overall system performance.

Maximizing Virtualization Capabilities with Intel Processor and Virtualization Technologies

Server uptime and server availability are not synonymous. A system can be up but not available, as in the case of a network outage. However, the reliability of your server hardware is still a fundamental aspect of keeping your vital IT workloads available. Many simple causes of server downtime can be addressed by using high-quality, redundant hardware (network interface cards, power supplies, etc.).

Because a single server can host many virtual machines, the reliability and performance of that server is even more important. Intel Xeon processors are designed to provide leading reliability, availability, and serviceability (RAS) for physical and virtual environments; and to help enable the full features and capabilities promised by a dynamic virtualized infrastructure.

Intel's Holistic Approach to Virtualization

As part of its commitment to providing robust virtualization solutions, Intel has adopted a holistic approach to virtualization. Intel Xeon processors deliver a hardware platform optimized for the best possible performance, reliability, and energy efficiency for virtualized environments.

Intel Virtualization Technology

At the core of the Intel virtualization strategy is Intel Virtualization Technology (Intel VT), which is embedded in Intel Xeon processors. Intel VT provides hardware assistance to virtualization software, allowing for reduced the size and complexity of the virtual machine manager (VMM) and enabling less expensive, more efficient, and more powerful virtualization solutions.

Specifically, Intel VT:

- Reduces the need for compute-intensive software translations between the guest and host operating systems in a virtual environment
- Reduces overhead by allowing the guest operating system to directly access CPU cycles
- Allows more virtual machines to run on a host server at near-native performance
- Supports 32-bit and 64-bit applications—as well as legacy operating systems—running on the same physical server
- Enhances overall server performance

Intel VT-x

The core component of the Intel VT road map, Intel VT-x is present in the processor and enables a simpler and more secure virtualization software platform. Intel VT-x:

- Reduces the size and complexity of the hypervisor software, improving its efficiency and enabling greater functionality
- Improves efficiency by reducing the need for virtualization software interventions through complex, compute-intensive software translations
- Enables greater functionality by allowing guest operating systems to run directly on the hardware, avoiding the need to modify them

Guest virtual machines run faster with Intel VT-x by allowing guest operating systems to run at their intended ring level. Intel VT-x also helps guest virtual machines boot faster.

Intel VT FlexPriority

With Intel VT FlexPriority, you can see even greater performance improvements. FlexPriority minimizes the impact of requests or “interrupts” from other devices or applications to the processor. It does this by using a special register to monitor the priority of tasks so that only interrupts with the highest priority get immediate attention.

Intel VT Extended Page Tables

The Intel VT Extended Page Tables (EPT) feature optimizes and accelerates the virtualization of physical memory. Memory virtualization allows virtual machines to see a virtual, contiguous

memory address space, instead of the non-contiguous underlying physical memory. This memory virtualization reduces overhead, improves virtual machine performance, and is accomplished via page tables, maintained in the guest virtual machine’s operating system. Prior to Extended Page Tables, these page tables could not be mapped directly to the physical memory; instead, it was mapped using additional layer of memory virtualization inside the VMM using shadow page tables.

Extended Page Tables accelerate this additional layer of memory virtualization with hardware assistance.

- EPT eliminates shadow page tables completely, resulting in reduced complexity and improved performance.
- EPT reduces the frequency of virtual machine exits to the VMM, improving performance and reliability.

All in all, with Intel VT you can achieve near-native performance of virtual workloads, optimize IT infrastructure, and improve server availability—all through the hardware level of a virtualization solution.

The greatly enhanced performance, energy efficiency, and reliability of Intel Xeon processor-based server systems make them the ideal choice for virtualization and consolidation projects or high-density deployments in power constrained data centers.

Intel Xeon 7400 series

The Intel® Xeon® processor 7400 series offers the industry’s highest virtualization performance so you can do more with less. With key platform innovations built in, the Intel Xeon processor 7400 series offers more headroom, reliability, and the highest expandability for virtualized solution environments.

Intel Xeon 5500 series

To gain the full value of virtualization technology, you need servers that are built to handle the heavy and ever-changing demands of a virtualized and consolidated computing environment. To help you get maximum benefits from virtualization, Intel has built a powerful physical platform with unique hardware-assist features to enhance the virtual data center.

The Intel® Xeon® processor 5500 Series, built on the new Intel® Microarchitecture (formerly codenamed “Nehalem”), is the first of this new generation of server platforms built with higher I/O bandwidth for higher virtualization performance plus multi-generation virtual machine migration for unparalleled flexibility in virtualized environments. In addition, Intel VT present in the Xeon 5500 series processors enhances native virtualization performance and reduces round-trip virtualization latency with hardware-enhanced technologies built into the Intel processor, chipset, and network adapter.

Intel Xeon 7400 series

Feature	Benefits
6-Core Processing	- Platform-compatible with Intel Xeon processor 7300 series for ease of migration and IT stability - Increased performance with 45 nm technology and increased headroom for multi-threaded applications and data-demanding applications - Enables improved virtualization performance, increasing system utilization
Enhanced Intel Core Microarchitecture	- Boosts performance on multiple applications / user environments and data-demanding workloads, while enabling denser data center deployments through improved performance per watt. 45 nm Hi-k process technology enables larger on-die cache for better performance - Reduced transistor gate leakage results in greater energy efficiency
16 MB of L3 Cache	- Increases efficiency of cache-to-core data transfers - Reduces latency by storing larger data sets closer to the processor, reducing the number of trips to main memory
Intel Virtualization Technology	- A suite of processor hardware enhancements that assists virtualization software to deliver more efficient virtualization solutions and greater capabilities, including 64-bit guest operating system support - Intel VT FlexMigration enables Intel Xeon processor 7400 series-based systems to be added to the existing virtualization pool with one, two, or 4+ socket Intel® Core™ microarchitecture-based servers
1066 MHz Dedicated High-Speed Interconnects (DHSI)	Enables increased throughput and bandwidth between each of the processors and the chipset
Intel 64 Architecture	Flexibility for 64-bit and 32-bit applications and operating systems

Intel Xeon 5500 series

Feature	Benefits
Intel Turbo Boost Technology	- Delivers performance as needed by adapting processor frequency to application needs - Saves energy in low-demand periods - Allows specific cores within each processor to operate above rated frequency within a set range, increasing frequency when needed to increase execution speed
Intel Hyper-Threading Technology	- Enables simultaneous multi-threading within each processor core - Allows up to two threads per core, or eight threads on a quad-core processor - Reduces computational latency, making optimal use of every cycle
Intel QuickPath Technology	- Delivers top performance for bandwidth-intensive applications - Integrates a memory controller into each microprocessor - Connects processors and other components with a new high-speed interconnect - Yields exceptional interconnect bandwidth to unleash the full performance of the next-generation Intel microarchitecture and future generations of Intel multi-core, dynamically scalable microprocessors

The Synergy between Hyper-V and Intel VT

Microsoft has continued its long-standing collaboration with Intel by maximizing the use of Hyper-V running on Intel Xeon processor-based servers with Intel VT. Hyper-V is designed to get the most from Intel VT to deliver greater performance and a lower TCO. This synergy between Hyper-V and Intel VT includes:

- Support for 24 cores, which means Hyper-V can take advantage of the performance enhancements of 6 core Intel® Xeon® 7400 processor-based servers
- Use of Intel's per-core power controls through the Demand-Based Switching capability for efficient power management
- Use of Intel VT FlexPriority to reduce the interrupt overhead and increase performance
- Support of Intel® Dynamic Power Manager, which allows IT to monitor and manage the power utilization of the servers
- Reduction in virtualization software overhead (and increased performance) through the use of:
 - Intel VT-x, which removes the needs for binary translations, thus reducing overhead and increasing performance
 - Intel VT for Directed I/O (VT-d), for reliability and security through device isolation and I/O performance with direct assignment (supported with Windows Server 2008 R2 Hyper-V)

Windows Server 2008 Hyper-V takes full advantage of Intel VT, delivering a more power-efficient performance and a lower total cost of ownership.

Windows Server 2008 Hyper-V takes advantage of Intel VT, delivering a more power-efficient performance and a lower total cost of ownership.

Putting It All Together: Combining Host and Guest Clustering for High Availability and Disaster Recovery

Host Clustering

You can host and failover virtual machines by making your physical host servers highly available. This is known as *host clustering*, and is a commonly recommended deployment. Host clustering allows you to separate each service or application and place them on individual virtual machines that are highly available. If a single virtual machine becomes unavailable, only the service hosted on that virtual machine

will fail over (assuming each virtual machine has its own LUN). In contrast, failure of a single virtual machine hosting multiple services would cause all those services to become unavailable.

For storage used by physical failover clusters, you can use iSCSI, Fibre Channel, or Serial Attached SCSI (SAS). When the physical host and virtual machine are in a running state, the administrator can gracefully fail the virtual machine over to another physical machine, and the end user should only need to wait a few seconds for the virtual machine to reconnect. In the background, host failover clustering will save the current running state of the virtual machine on the shared storage, stop it on the first node, and start it on the new node.

Failover clustering of the physical host also monitors the health of the virtual machines, so if the virtual operating system crashes, hangs, or blue-screens, it can be automatically restarted; however, the end user would need to wait a little longer for the virtual machine state to load and start.

Intel Xeon processor-based servers enhance clustering scenarios with their leading reliability, accessibility, and serviceability (RAS) features. While clustering protects the workloads running on each node, Intel Xeon processor-based servers help keep the reduced number of physical host servers up and running. Providing a highly reliable and available hardware platform becomes a much greater imperative for businesses seeking to harness the benefits of virtualization.

The exceptional RAS features of Intel Xeon processors provide this platform by reducing the cost and complexity of high-availability hardware solutions while improving recovery speed and reliability. These features include improved Error Correcting Code (ECC) coverage for system bus and cache, new memory mirroring, fully buffered DIMM technology, and hot-pluggable component support.

Guest Clustering

While host clustering can monitor the state of the virtual operating system, it does not monitor the health of applications inside the virtual machines. The solution for this problem is *guest clustering*.

Guest clustering enables highly available services and applications in the virtual layer and is fully supported in Windows Server 2008. Guest clustering is done by installing failover clustering on several virtual machines, then clustering them as if they were physical nodes. All of these virtual machines can use the same shared iSCSI storage (virtual machines do not support other methods of shared storage), which will store the state of the running applications within the virtual machine.

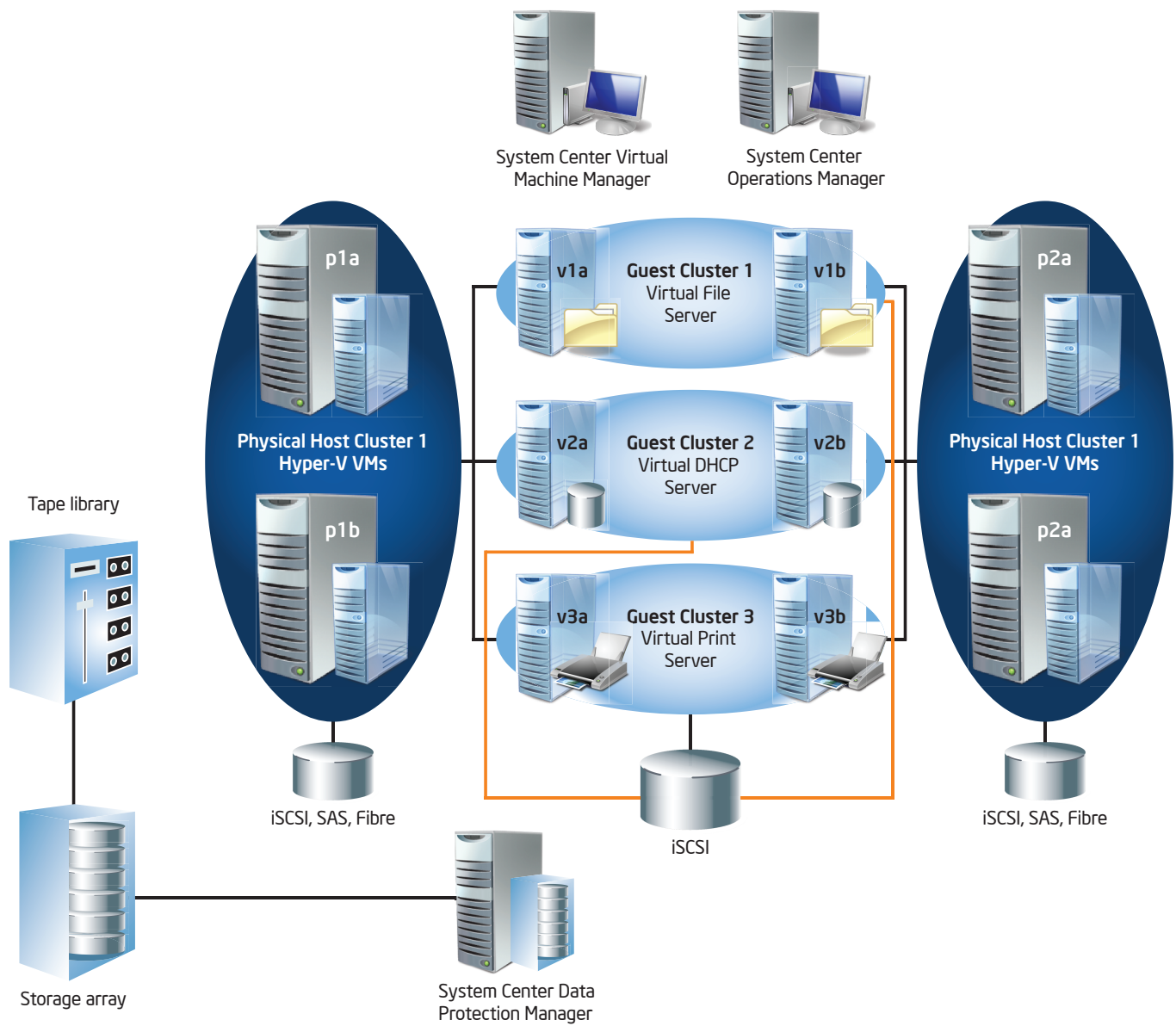
Guest clustering with Hyper-V is enhanced by using Intel VT. With simultaneous support for 16-bit, 32-bit, and 64-bit guest operating systems, Intel VT helps ensure that all of an organization's virtual machines can be protected with guest clustering. Intel VT also helps guest virtual machines transition quickly from powered-off to powered-on states, minimizing downtime during Quick Migration.

Making your services highly available in the virtual layer allows you monitor their health and fail them over if the virtual machine is experiencing downtime. Combining guest clustering and host clustering enables you to provide high availability for both physical and virtual environments, right down to the application or service level.

Microsoft System Center

System Center Data Protection Manager (DPM) enables Disaster Recovery for your highly available virtual infrastructure. DPM can be used to back up both the physical and virtual infrastructures. The DPM agent can be installed on both the physical and virtual servers to enable live backups of all servers, thus minimizing downtime and user interruption. In the event that shared storage fails and data is lost, DPM can be used to quickly restore both physical and virtual servers as well as the application data contained on all servers in the design.

To keep tabs on the state of your various physical and virtual machines, System Center Operations Manager provides end-to-end service monitoring in your enterprise IT environment. Operations Manager provides an easy-to-use monitoring environment that monitors thousands of servers, applications, and clients to provide a comprehensive view of the health of your organization's IT environment. This view of service health is a key to a rapid, agile response to events that may impact the normal running of business and ultimately cost the organization money.



Example Architecture

Figure 1 (above) depicts the end-to-end high availability solution that combines both host and guest failover clustering.

- Physical Node p1a and p1b are servers powered by Intel Xeon processors with Intel VT that form a physical host cluster. They host virtual servers v1a, v2a, and v3a. These virtual servers fail over between p1a and p1b, allowing high availability at the physical layer.
- Physical Node p2a and p2b are also Intel VT-enabled Xeon processor-based servers that form a physical host cluster. They host virtual servers v1b, v2b, and v3b. These virtual servers fail over between p2a and p2b, allowing high availability at the physical layer.
- Virtual nodes v1a and v1b are clustered in the virtual layer.
- Virtual nodes v2a and v2b are clustered in the virtual layer.
- Virtual nodes v3a and v3b are clustered in the virtual layer.
- Applications are now made highly available in the virtual layer, and they can fail over, while the virtual machines are made highly available at the physical layer, and they can fail over. This allows high availability for both the physical machines and the virtual machines.

Combining Hyper-V with Windows Server 2008 failover clustering, System Center Operations Manager, System Center Virtual Machine Manager and System Center Data Protection Manager, supported by Intel Xeon processors, enables the dynamic data center to achieve maximum uptime. By combining your guest and host clustering, you can create high availability in both your physical and virtual layers.

Summary of Virtualization Strategies and Benefits

Server virtualization at its best comes from a complementary approach that takes advantage of both hardware and software strengths. As key players in the field of virtualization, Intel and Microsoft have focused the power of their long-standing partnership to optimize the benefits that hardware assistance offers in virtual environments. The right combination of hardware and software provides businesses with robust management capabilities for their virtual and physical machines, resulting in the following time and cost-savings benefits:

- Optimized IT infrastructure
- Lower total cost of ownership
- Improved flexibility
- Higher reliability

Together, Intel and Microsoft offer built-in hardware and software virtualization technologies. Intel Xeon processors are designed to support virtual environments through Intel VT. Intel VT provides hardware assistance to the virtual machine manager, reducing software overhead associated with translating processor requests. Thus, Intel VT forms a rock-solid hardware foundation for running and managing virtual servers. Hyper-V delivers a robust hypervisor supported by advanced management tools found in System Center Virtual Machine Manager—both designed to capitalize on Intel VT. IT administrators can use these tools to deploy and manage multiple servers, both physical and virtual, as a single pool of resources, moving and balancing workloads among them without service interruption. Together, these advanced hardware and software technologies create a powerful solution for virtual and physical machine management. IT infrastructures that take advantage of these complementary technologies help make organizations more agile and IT operations more robust and efficient.

Related Links

Intel Xeon Processors

www.intel.com/products/server/processors/index.htm?iid=process+5300

Intel Server Virtualization

www.intel.com/business/technologies/virtualization.htm?iid=servproc+rhc_virtualization

Intel Xeon Processor 5500 series and Microsoft Solutions

<http://www.intelalliance.com/microsoft/intel-xeon-5500.html>

TechNet Webcast: Increase IT Flexibility and Responsiveness Through Hardware-Assisted Virtualization (Level 300)

<http://go.microsoft.com/fwlink/?LinkId=109811>

Next-Generation Intel Microarchitecture

www.intel.com/technology/architecture-silicon/next-gen/?iid=search

Hyper-V: A Key Feature of Windows Server 2008

www.microsoft.com/windowsserver2008/virtualization/default.aspx

Hyper-V: Frequently Asked Questions

www.microsoft.com/windowsserver2008/virtualization/faq.aspx

Windows System Center Virtual Machine Manager

www.microsoft.com/systemcenter/scvmm/default.aspx

TechNet Webcast: Managing Virtualization with System Center Virtual Machine Manager 2007 (Level 200)

<http://go.microsoft.com/fwlink/?LinkId=109844>

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